This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

## Device Specifications

Device Type: Audio Processor
Manufacturer: Cambridge Audio
Firmware Version: N/A
Model(s): Qt 300, Qt 600, Qt 100

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scripter Version |
|---|---|
| 3.13.0000-b006 | 2.13.0 |

## Version History

| Module Version | Date | Notes |
|---|---|---|
| 1_2_1_0 | 8/25/2021 | • Updated communication Sheet regarding Multi-Connection<br>• Added ‘Type’ parameter to Input A Output level and Input B Output level commands.<br>• Fixed module support for model Qt 600. |
| 1_1_1_0 | 9/3/2019 | Initial Version |

## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  `Example: InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  `Example: InterfaceName.connectionCounter = 5`

- Recommended command pacing for this module is 0.4 seconds or higher when incrementing or decrementing the various settings. Using anything lower than 0.4 seconds will cause the device to freeze.

## Supported Class and Example

**EthernetClass**

```
    InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='Qt 300')
```

## Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

| | | | |
|---|---|---|---|
| Command<br>**AutoRamping** | Value<br>-20 to 0 in steps of 1 dB | | |
| Qualifier Key<br>**'Zone'** | Qualifier Value<br>'1'[1] | Qualifier Value<br>'1' – '3'[2] | Qualifier Value<br>'1' – '6'[3] |
| `# AutoRamping example`<br>`InterfaceName.Set('AutoRamping', 0, {'Zone': '1'})` | | | |
| Command<br>**InputAOutputLevel** | Value<br>0 to 30 in steps of 1 dB | | |
| Qualifier Key<br>**'Type'** | Qualifier Value<br>'Max' | Qualifier Value<br>'Min' | |
| Qualifier Key<br>**'Zone'** | Qualifier Value<br>'1'[1] | Qualifier Value<br>'1' – '3'[2] | Qualifier Value<br>'1' – '6'[3] |
| `# InputAOutputLevel example`<br>`InterfaceName.Set('InputAOutputLevel', 30, {'Type': 'Max', 'Zone': '1'})` | | | |
| Command<br>**InputBOutputLevel** | Value<br>0 to 30 in steps of 1 dB | | |
| Qualifier Key<br>**'Type'** | Qualifier Value<br>'Max' | Qualifier Value<br>'Min' | |
| Qualifier Key<br>**'Zone'** | Qualifier Value<br>'1'[1] | Qualifier Value<br>'1' – '3'[2] | Qualifier Value<br>'1' – '6'[3] |
| `# InputBOutputLevel example`<br>`InterfaceName.Set('InputBOutputLevel', 30, {'Type': 'Max', 'Zone': '1'})` | | | |
| Command<br>**MaskingLevel** | Value<br>0 to 30 in steps of 0.5 dB | | |
| Qualifier Key<br>**'Type'** | Qualifier Value<br>'Max' | Qualifier Value<br>'Min' | |
| Qualifier Key<br>**'Zone'** | Qualifier Value<br>'1'[1] | Qualifier Value<br>'1' – '3'[2] | Qualifier Value<br>'1' – '6'[3] |
| `# MaskingLevel example`<br>`InterfaceName.Set('MaskingLevel', 30, {'Type': 'Max', 'Zone': '1'})` | | | |
| Command<br>**Mute** | Value<br>'On' | Value<br>'Off' | |
| Qualifier Key<br>**'Zone'** | Qualifier Value<br>'1'[1] | Qualifier Value<br>'1' – '3'[2] | Qualifier Value<br>'1' – '6'[3] |
| `# Mute example`<br>`InterfaceName.Set('Mute', 'On', {'Zone': '1'})` | | | |

[1] Only supported for model Qt 100.
[2] Only supported on model Qt 300.
[3] Only supported for model Qt 600.

## Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
        InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
        Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
        InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
FeedbackHandler)
        FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
        InterfaceName.Update(Command)
        Value = InterfaceName.ReadStatus(Command)
        InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
        FeedbackHandler will be called when any qualifier gets a new status.
```

| Command **AutoRamping** | Value -20 to 0 in steps of 1 dB | | |
|---|---|---|---|
| Qualifier Key **'Zone'** | Qualifier Value '1' [1] | Qualifier Value '1' – '3' [2] | Qualifier Value '1' – '6' [3] |

```
# AutoRamping example
InterfaceName.Update('AutoRamping', {'Zone': '1'})
Value = InterfaceName.ReadStatus('AutoRamping', {'Zone': '1'})
InterfaceName.SubscribeStatus('AutoRamping', None, FeedbackHandler)
```

| Command **ConnectionStatus** | Value 'Connected' | Value 'Disconnected' |
|---|---|---|

```
# ConnectionStatus example
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

| Command **ExecutiveMode** | Value 'On' | Value 'Off' |
|---|---|---|

```
# ExecutiveMode example
InterfaceName.Update('ExecutiveMode')
Value = InterfaceName.ReadStatus('ExecutiveMode')
InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)
```

| Command **InputAOutputLevel** | Value 0 to 30 in steps of 1 dB | | |
|---|---|---|---|
| Qualifier Key **'Type'** | Qualifier Value 'Max' | Qualifier Value 'Min' | |
| Qualifier Key **'Zone'** | Qualifier Value '1' [1] | Qualifier Value '1' – '3' [2] | Qualifier Value '1' – '6' [3] |

```
# InputAOutputLevel example
InterfaceName.Update('InputAOutputLevel', {'Type': 'Max', 'Zone': '1'})
Value = InterfaceName.ReadStatus('InputAOutputLevel', {'Type': 'Max', 'Zone': '1'})
InterfaceName.SubscribeStatus('InputAOutputLevel', None, FeedbackHandler)
```

| Command **InputBOutputLevel** | Value 0 to 30 in steps of 1 dB | | |
|---|---|---|---|
| Qualifier Key **'Type'** | Qualifier Value 'Max' | Qualifier Value 'Min' | |
| Qualifier Key **'Zone'** | Qualifier Value '1' [1] | Qualifier Value '1' – '3' [2] | Qualifier Value '1' – '6' [3] |

```
# InputBOutputLevel example
InterfaceName.Update('InputBOutputLevel', {'Type': 'Max', 'Zone': '1'})
Value = InterfaceName.ReadStatus('InputBOutputLevel', {'Type': 'Max', 'Zone': '1'})
```

```
InterfaceName.SubscribeStatus('InputBOutputLevel', None, FeedbackHandler)
```

| Command | Value | | |
|---|---|---|---|
| **MaskingLevel** | 0 to 30 in steps of 0.5 dB | | |
| Qualifier Key **'Type'** | Qualifier Value 'Max' | Qualifier Value 'Min' | |
| Qualifier Key **'Zone'** | Qualifier Value '1' [1] | Qualifier Value '1' – '3' [2] | Qualifier Value '1' – '6' [3] |

```
# MaskingLevel example
InterfaceName.Update('MaskingLevel', {'Type': 'Max', 'Zone': '1'})
Value = InterfaceName.ReadStatus('MaskingLevel', {'Type': 'Max', 'Zone': '1'})
InterfaceName.SubscribeStatus('MaskingLevel', None, FeedbackHandler)
```

| Command | Value | Value | |
|---|---|---|---|
| **Mute** | 'On' | 'Off' | |
| Qualifier Key **'Zone'** | Qualifier Value '1' [1] | Qualifier Value '1' – '3' [2] | Qualifier Value '1' – '6' [3] |

```
# Mute example
InterfaceName.Update('Mute', {'Zone': '1'})
Value = InterfaceName.ReadStatus('Mute', {'Zone': '1'})
InterfaceName.SubscribeStatus('Mute', None, FeedbackHandler)
```

[1] Only supported for model Qt 100.

[2] Only supported on model Qt 300.

[3] Only supported for model Qt 600.

## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface.

| | |
|---|---|
| **Port Type:** | Ethernet (TCP) |
| **Default Port:** | 23 |
| **Logon Credentials Supported:** | No |
| **Multi-Connection Capabilities:** | No |
| **Port Changeability:** | Yes |

## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication.

## Notes for the Device

## Appendix A. Set Commands

| | |
|---|---|
| **Auto Ramping 0 Zone 1** | ZOSET,AUTO0=0\x0D |
| **Auto Ramping 0 Zone 3** | ZOSET,AUTO2=0\x0D |
| **Auto Ramping 0 Zone 6** | ZOSET,AUTO5=0\x0D |
| **Auto Ramping -20 Zone 1** | ZOSET,AUTO0=-20\x0D |
| **Auto Ramping -20 Zone 3** | ZOSET,AUTO2=-20\x0D |
| **Auto Ramping -20 Zone 6** | ZOSET,AUTO5=-20\x0D |
| **Input A Output Level 0 Type Max Zone 1** | ZOSET,INAM0=0\x0D |
| **Input A Output Level 0 Type Max Zone 3** | ZOSET,INAM2=0\x0D |
| **Input A Output Level 0 Type Max Zone 6** | ZOSET,INAM5=0\x0D |
| **Input A Output Level 0 Type Min Zone 1** | ZOSET,INAI0=0\x0D |
| **Input A Output Level 0 Type Min Zone 3** | ZOSET,INAI2=0\x0D |
| **Input A Output Level 0 Type Min Zone 6** | ZOSET,INAI5=0\x0D |
| **Input A Output Level 30 Type Max Zone 1** | ZOSET,INAM0=30\x0D |
| **Input A Output Level 30 Type Max Zone 3** | ZOSET,INAM2=30\x0D |
| **Input A Output Level 30 Type Max Zone 6** | ZOSET,INAM5=30\x0D |
| **Input A Output Level 30 Type Min Zone 1** | ZOSET,INAI0=30\x0D |
| **Input A Output Level 30 Type Min Zone 3** | ZOSET,INAI2=30\x0D |
| **Input A Output Level 30 Type Min Zone 6** | ZOSET,INAI5=30\x0D |
| **Input B Output Level 0 Type Max Zone 1** | ZOSET,INBM0=0\x0D |
| **Input B Output Level 0 Type Max Zone 3** | ZOSET,INBM2=0\x0D |
| **Input B Output Level 0 Type Max Zone 6** | ZOSET,INBM5=0\x0D |
| **Input B Output Level 0 Type Min Zone 1** | ZOSET,INBI0=0\x0D |
| **Input B Output Level 0 Type Min Zone 3** | ZOSET,INBI2=0\x0D |
| **Input B Output Level 0 Type Min Zone 6** | ZOSET,INBI5=0\x0D |
| **Input B Output Level 30 Type Max Zone 1** | ZOSET,INBM0=30\x0D |
| **Input B Output Level 30 Type Max Zone 3** | ZOSET,INBM2=30\x0D |
| **Input B Output Level 30 Type Max Zone 6** | ZOSET,INBM5=30\x0D |
| **Input B Output Level 30 Type Min Zone 1** | ZOSET,INBI0=30\x0D |
| **Input B Output Level 30 Type Min Zone 3** | ZOSET,INBI2=30\x0D |
| **Input B Output Level 30 Type Min Zone 6** | ZOSET,INBI5=30\x0D |
| **Masking Level 0 Type Max Zone 1** | ZOSET,MMAX0=0\x0D |
| **Masking Level 0 Type Max Zone 3** | ZOSET,MMAX2=0\x0D |
| **Masking Level 0 Type Max Zone 6** | ZOSET,MMAX5=0\x0D |
| **Masking Level 0 Type Min Zone 1** | ZOSET,MMIN0=0\x0D |
| **Masking Level 0 Type Min Zone 3** | ZOSET,MMIN2=0\x0D |
| **Masking Level 0 Type Min Zone 6** | ZOSET,MMIN5=0\x0D |
| **Masking Level 30 Type Max Zone 1** | ZOSET,MMAX0=60\x0D |
| **Masking Level 30 Type Max Zone 3** | ZOSET,MMAX2=60\x0D |
| **Masking Level 30 Type Max Zone 6** | ZOSET,MMAX5=60\x0D |
| **Masking Level 30 Type Min Zone 1** | ZOSET,MMIN0=60\x0D |
| **Masking Level 30 Type Min Zone 3** | ZOSET,MMIN2=60\x0D |
| **Masking Level 30 Type Min Zone 6** | ZOSET,MMIN5=60\x0D |

| | |
|---|---|
| **Mute Off Zone 1** | ZOSET,MUTE0=0\x0D |
| **Mute Off Zone 3** | ZOSET,MUTE2=0\x0D |
| **Mute Off Zone 6** | ZOSET,MUTE5=0\x0D |
| **Mute On Zone 1** | ZOSET,MUTE0=1\x0D |
| **Mute On Zone 3** | ZOSET,MUTE2=1\x0D |
| **Mute On Zone 6** | ZOSET,MUTE5=1\x0D |

## Appendix B. Update Commands

| | |
|---|---|
| **Auto Ramping Zone 1** | ZOGET,AUTO0\x0D |
| **Auto Ramping Zone 3** | ZOGET,AUTO2\x0D |
| **Auto Ramping Zone 6** | ZOGET,AUTO5\x0D |
| **Executive Mode** | CSGET,LOCK\x0D |
| **Input A Output Level Type Max Zone 1** | ZOGET,INAM0\x0D |
| **Input A Output Level Type Max Zone 3** | ZOGET,INAM2\x0D |
| **Input A Output Level Type Max Zone 6** | ZOGET,INAM5\x0D |
| **Input A Output Level Type Min Zone 1** | ZOGET,INAI0\x0D |
| **Input A Output Level Type Min Zone 3** | ZOGET,INAI2\x0D |
| **Input A Output Level Type Min Zone 6** | ZOGET,INAI5\x0D |
| **Input B Output Level Type Max Zone 1** | ZOGET,INBM0\x0D |
| **Input B Output Level Type Max Zone 3** | ZOGET,INBM2\x0D |
| **Input B Output Level Type Max Zone 6** | ZOGET,INBM5\x0D |
| **Input B Output Level Type Min Zone 1** | ZOGET,INBI0\x0D |
| **Input B Output Level Type Min Zone 3** | ZOGET,INBI2\x0D |
| **Input B Output Level Type Min Zone 6** | ZOGET,INBI5\x0D |
| **Masking Level Type Max Zone 1** | ZOGET,MMAX0\x0D |
| **Masking Level Type Max Zone 3** | ZOGET,MMAX2\x0D |
| **Masking Level Type Max Zone 6** | ZOGET,MMAX5\x0D |
| **Masking Level Type Min Zone 1** | ZOGET,MMIN0\x0D |
| **Masking Level Type Min Zone 3** | ZOGET,MMIN2\x0D |
| **Masking Level Type Min Zone 6** | ZOGET,MMIN5\x0D |
| **Mute Zone 1** | ZOGET,MUTE0\x0D |
| **Mute Zone 3** | ZOGET,MUTE2\x0D |
| **Mute Zone 6** | ZOGET,MUTE5\x0D |